



ACADEMIC PROJECT REVIEW • MILESTONE 1

# SkillVerse

AI-Powered Peer Learning, Dynamic Skill Assessment & Career Guidance  
Platform

PROJECT STATUS

**Phase 1 (30% Scope) Complete**

ARCHITECTURE

**Next.js 16 + Neon PostgreSQL**

AI ENGINE

**Groq LLaMA-3.3 70B**

PRESENTER / LEAD

**Sreepathy (Team SkillVerse)**

# Phase 1 Milestone: 30% Architecture Scope

Establishing the core foundation: user roles, dynamic AI assessment, and mentor accreditation.

## 30%

### PROJECT SCOPE COMPLETED

100% of defined Phase 1 deliverables fully implemented and live.

## 3 Tiers

### AI DIFFICULTY LEVELS

Simple, Medium, and Hard dynamic tests tailored to candidate profile.

## 4 Badges

### MENTOR ACCREDITATION LEVELS

Master (5.0★), Senior (4.8★), Associate (4.5★), & Pending.

## 0 Defect

### QUALITY ASSURANCE

0 TypeScript errors, 0 ESLint warnings, 25 compiled routes.

### 🎯 Core Problem Solved in Phase 1

- ▶ **Eliminates Static Tests:** Generic multiple-choice tests fail to evaluate actual student profile skills. SkillVerse generates dynamic questions for candidate's exact technologies.
- ▶ **Solves Fake Mentors:** Anyone can claim to be a mentor online. SkillVerse requires an automated AI Technical & Pedagogy Accreditation test before unlocking badges and ads.
- ▶ **Clutter-Free UX:** Strict separation ensures students never see mentor admin tools, and mentors never get trapped in student beginner roadmaps.

### ⚡ Key Engineering Accomplishments

- ▶ Implemented stateless JWT authentication with bcrypt password security and role gating.
- ▶ Deployed Neon serverless PostgreSQL with Prisma ORM transactional integrity.
- ▶ Integrated Groq Cloud API with LLaMA-3.3 70B in strict JSON schema mode.
- ▶ Engineered resilient deterministic fallbacks ensuring 100% platform uptime.

# Project Completion Matrix: 30% vs Remaining 70%

Clear demarcation of completed foundational bricks versus upcoming interactive modules.

| MODULE NAME                          | TARGET SCOPE | KEY DELIVERABLES                                                                              | STATUS             |
|--------------------------------------|--------------|-----------------------------------------------------------------------------------------------|--------------------|
| Brick 1: Profile System & XP         | 10%          | Student/Mentor profile models, skills tag cloud, career goal, education, XP points.           | DONE (100%)        |
| Brick 2: 3-Tier AI Assessment        | 10%          | Profile skill extraction, Simple/Medium/Hard tiers, Groq LLM JSON pipeline, verified badges.  | DONE (100%)        |
| Brick 3: Mentor Accreditation & Ads  | 10%          | Pedagogy evaluation, Master/Senior/Associate levels, 5.0★ rating, Mentor Studio ad publisher. | DONE (100%)        |
| Brick 4: Live 1-on-1 Rooms & Chat    | 25%          | WebRTC audio/video call room, synchronized code editor, real-time WebSocket chat.             | PHASE 2 (UPCOMING) |
| Brick 5: Payment & Slot Booking      | 20%          | Stripe/Razorpay gateway for paid mentorship, Google Calendar slot booking & reservations.     | PHASE 2 (UPCOMING) |
| Brick 6: AI Interviewer & Embeddings | 25%          | Voice mock interview simulator (Whisper), pgvector semantic peer matchmaking, certificates.   | PHASE 3 (UPCOMING) |

# Profile Architecture & Experience Points (XP)

Centralized data identity engine supporting student learning and mentor credibility.

## Student Profile Attributes

- ▶ **Verified Skills:** Dynamic string array linked to assessment verification.
- ▶ **Career Vision:** Target role (e.g. Full-Stack Engineer, ML Engineer).
- ▶ **Education & Interests:** College, degree, domain interests for peer matching.
- ▶ **Social Portfolio:** GitHub and LinkedIn integration.

## Mentor Profile Attributes

- ▶ **Official Mentor Level:** Persisted AI-decided tier ('Master', 'Senior', 'Associate').
- ▶ **Public Star Rating:** Computed from accreditation score ('5.0★', '4.8★', '4.5★').
- ▶ **Mentoring Domains:** Verified technical specialties.
- ▶ **Ad Offerings:** Relation to published skill posts and student booking requests.

## XP Gamification Engine

- ▶ **Profile Completion:** +20 XP awarded upon completing all credential fields.
- ▶ **Skill Verification:** +10 XP for passing assessments ( $\geq 60\%$ ).
- ▶ **Mentor Accreditation:** +15 XP for passing pedagogical accreditation.
- ▶ **Roadmap Milestones:** +5 XP per completed milestone waypoint.

## Server-Side Validation & Security (Zod + Prisma)

All profile updates pass through strict Zod schemas on the server boundary. Sensitive fields like passwords and auth tokens are stripped from profile queries. Relational joins ensure atomic updates with `prisma.$transaction`.

# Dynamic AI Assessment Engine & 3 Difficulty Tiers

Adaptive 5-question technical evaluation calibrated to the user's selected difficulty.

## Simple Tier

### FOUNDATIONS

#### Syntax & Core Definitions

- ▶ Core language fundamentals & keywords.
- ▶ Variable mutability (const vs let).
- ▶ Semantic HTML5 & CSS Flexbox alignment.
- ▶ Zero trick questions; builds learner confidence.

## Medium Tier

### APPLIED SYSTEMS

#### Real-World Implementation

- ▶ Asynchronous execution & Node.js event loop.
- ▶ Hash table time complexity & B-Tree index lookup.
- ▶ React hook memoization (useCallback/useEffect).
- ▶ Restful HTTP idempotency (GET/PUT/DELETE).

## Hard Tier

### ADVANCED INTERNALS

#### Architecture & Concurrency

- ▶ Raft distributed consensus split-vote timers.
- ▶ PostgreSQL MVCC snapshot concurrency.
- ▶ Cache stampede & check-then-act race conditions.
- ▶ React 19 Fiber streaming SSR error boundaries.

## ✦ Profile-Driven Skill Auto-Sync

When a student enters the assessment center, the system automatically pulls their verified skills from their profile as primary selectable pills. Clicking any skill instantly initializes a tailored 5-question test at the desired difficulty tier.

# Groq LLaMA-3.3 70B AI Integration Pipeline

Ultra-low latency LLM inference with strict JSON schema contracts and offline resilience.

```
// System Contract: Strict JSON Enforcement POST
https://api.groq.com/openai/v1/chat/completions { "model": "llama-3.3-70b-versatile", "response_format": { "type": "json_object" }, "temperature": 0.2, "messages": [ { "role": "system", "content": "Output strictly valid JSON..." }, { "role": "user", "content": "Generate 5 ${difficulty} questions..." } ] }
```

## Key Engineering Highlights

- ▶ **High-Speed LPU Inference:** Groq's Language Processing Units generate 70B token streams at ~400 tokens/sec, eliminating long spinner waits.
- ▶ **Zod Runtime Validation:** Every LLM response is validated against `groqQuizResponseSchema` before touching the frontend, preventing malformed UI renders.
- ▶ **Profile Context Injection:** Candidate headline, existing skills, and education are injected into the prompt so questions match their academic background.
- ▶ **Automated Fast Fallback:** If the 70B model encounters rate limits, the client retries on `llama-3.1-8b-instant` automatically.

## Deterministic Question Engine (Zero-Downtime Guarantee)

If the user has not configured their API key or network connectivity drops, the backend automatically engages pre-compiled, topic-specific question banks for Python, React, TypeScript, and Architecture across all 3 tiers. The application NEVER crashes.

# Mentor Competency Accreditation & Star Ratings

Automated assessment determining mentor tier badges, public star ratings, and credibility.



## Master Mentor

**Score  $\geq 90\%$  • 5.0 ★ Public Rating**

Tier 1: Principal Architect & Capstone Lead

Certified to lead complex system design evaluations, capstone code reviews, and architectural masterclasses.



## Senior Mentor

**Score 75% - 89% • 4.8 ★ Public Rating**

Tier 2: Advanced Systems & Code Reviewer

Certified for deep-dive code reviews, full-stack debugging, refactoring anti-patterns, and interview prep.



## Associate Mentor

**Score 60% - 74% • 4.5 ★ Public Rating**

Tier 3: Core Foundations Guide

Certified for foundational tutoring, syntax coaching, beginner debugging assistance, and concept walkthroughs.



## Pedagogical Testing: Evaluating "How to Teach", Not Just "How to Code"

Mentor questions specifically evaluate code review empathy (constructive PR feedback), diagnosing student mental models (explaining async via analogies), detecting security flaws (SQL injection / IDOR), and guiding performance without over-engineering.

# Strict Role Separation: No Non-Logical Options

Clear cognitive separation between Student learners and accredited Mentors.



## Student Dedicated Experience

- ▶ **Assessment Center:** Shows skill verification pills, 3 difficulty tiers, and student verification badges.
- ▶ **Automated Route Redirection:** After submitting, students see a "Calibrating Career Path..." route modal directing them to their calibrated career roadmap.
- ▶ **Completely Hidden:** Students never see mentor accreditation deciders, tier level calculations, or "Post Skill Ad" buttons.
- ▶ **Navigation:** Header displays Find Mentors, AI Skill Assessment, and Career Roadmap.



## Mentor Dedicated Experience

- ▶ **Accreditation Portal:** Shows current Mentor Level, Star Rating badge, and pedagogical evaluation tiers (Associate, Senior, Master).
- ▶ **Post-Test Actions:** Directly navigates to **Mentor Studio** and Mentor Profile.
- ▶ **Completely Hidden:** Mentors never see student beginner roadmaps, waypoint redirects, or non-applicable curriculum buttons.
- ▶ **Navigation:** Header displays Mentor Studio (+ Post Ad),  AI Accreditation, and Find Peers.



## Interactive Profile Skill Badges

On both Student and Mentor profiles, skills are rendered as active interactive chips. Clicking any skill badge directly launches a customized assessment in that specific technology, eliminating manual re-typing.

# Mentor Studio & Student Discovery Hub

Empowering accredited mentors to post skill offerings and let students discover them.

## Skill Ad Publishing Engine

- ▶ Offering Title & Description.
- ▶ Domain Category (Frontend, Backend, AI, System Design).
- ▶ Flexible Pricing: **100% Free** or **Paid Session (INR/USD)**.
- ▶ Availability Slot Schedule (e.g. Weekends 10 AM - 2 PM).

## Real-Time Studio Dashboard

- ▶ **Active Ads Counter:** Tracks live published offerings.
- ▶ **Inquiries Tracker:** Aggregates student session bookings.
- ▶ **Accreditation Badge:** Showcases verified level & rating.
- ▶ **Status Toggles:** One-click ACTIVE vs PAUSED toggle.

## Public Search Discovery (`/search`)

- ▶ Student directory indexing all active mentor skill ads.
- ▶ Real-time filtering by category and search keyword.
- ▶ Displays verified mentor rating badges (`5.0★`, `Master Mentor`).
- ▶ Direct booking trigger opening student inquiry modals.

## Persistence & Data Flow

Every published ad creates a record in the Post model linked to the mentor's user ID with cascade integrity. First-time ad publishing awards +10 XP to the mentor's account.

# Dynamic Career Roadmapping & Calibrated Routing

Visual learning navigation from current verified baseline to dream engineering roles.

## Seamless Assessment-to-Roadmap Calibration

- ▶ **Direct Routing:** When a student scores  $\geq 60\%$  in an assessment, the system automatically routes them to /roadmap?fromAssessment=1&score=....
- ▶ **Waypoint Verification:** Baseline milestones matching the tested skill are automatically marked as verified and completed.
- ▶ **Target Pathways:** Curated tracks for Full-Stack Software Engineer, AI/ML Engineer, and UI/UX Product Designer.

## Custom AI Pathway Generator

- ▶ **Dream Role Input:** Students can type any specialized role (e.g., Cloud Security Architect, Quant Developer).
- ▶ **Groq AI Generation:** Generates tailored milestones: Baseline, Skill Gap, Recommended, Capstone, and Target Role.
- ▶ **Gap Analysis:** Compares student's current profile skills against target role requirements and highlights missing competencies.

## Recommended Mentor Matching

Each roadmap milestone highlights recommended accredited mentors specializing in that specific skill gap, allowing students to book 1-on-1 sessions directly from their learning path.

# Technologies Used in Phase 1 ("What I Used in This")

Complete inventory of frameworks, libraries, databases, and AI tooling deployed.

## Frontend

- ▶ **Next.js 16:** Turbopack App Router, Server Components.
- ▶ **React 19:** Modern hooks, state colocation.
- ▶ **TypeScript 5.9:** Strict type safety.
- ▶ **Tailwind CSS:** Curated dark/light design system.

## Backend & Auth

- ▶ **Next.js Route Handlers:** RESTful API routes.
- ▶ **Jose JWT:** Cryptographic stateless auth.
- ▶ **BcryptJS:** Salted password hashing.
- ▶ **Zod:** Schema boundary validation.

## Database

- ▶ **Neon Serverless:** PostgreSQL cloud database.
- ▶ **Prisma ORM 6:** Schema modeling & migrations.
- ▶ **Transactions:** `$transaction` for atomicity.
- ▶ **Connection Pool:** SSL-secured pooling.

## AI & Tooling

- ▶ **Groq Cloud API:** LPU acceleration.
- ▶ **LLaMA-3.3 70B:** Versatile reasoning.
- ▶ **LLaMA-3.1 8B:** High-speed fallback.
- ▶ **ESLint & Turbopack:** Code verification.

## ★ Architectural Principle: Feature-Focused & Minimal Overhead

Zero unnecessary third-party abstractions. Used native Next.js features, standard React 19 paradigms, and type-safe Prisma queries to maintain high execution velocity and maintainability.

# Prisma Relational Database Architecture

Normalized relational schema deployed on Neon PostgreSQL.

```
model User { id String @id @default(cuid()) email String @unique
  role Role @default(STUDENT) xp Int @default(0) completedActivities
  String[] @default([]) profile Profile? assessments
  SkillAssessment[] posts Post[] @relation("MentorPosts") bookings
  Booking[] } model Profile { id String @id @default(cuid()) userId
  String @unique user User @relation(fields: [userId], references:
  [id]) skills String[] @default([]) mentorLevel String? mentorScore
  Float? careerGoal String? }
```

```
model Post { id String @id @default(cuid()) userId String user
  User @relation("MentorPosts", fields: [userId], references: [id])
  skillName String category String pricingType PricingType
  @default(FREE) priceAmount Float? status PostStatus
  @default(ACTIVE) bookings Booking[] } model SkillAssessment { id
  String @id @default(cuid()) userId String user User
  @relation(fields: [userId], references: [id]) skillName String
  score Float isVerified Boolean @default(false) feedback String? }
```

# Future Roadmap ("What Will Use in Future")

Planned technologies and milestones for Phase 2 (Interaction) and Phase 3 (Intelligence).

## Phase 2: Live Interaction & Booking (30% → 65%)

- ▶ **WebRTC 1-on-1 Video/Audio Rooms:** Deploy **LiveKit / Agora SDK** for in-browser video calls with a synchronized collaborative code editor.
- ▶ **Real-Time Messaging:** Integrate **WebSockets / Socket.io** for instant chat, direct inquiries, and session reminder notifications.
- ▶ **Payment Gateway:** Integrate **Stripe Connect / Razorpay** for secure escrow payments on paid mentorship sessions.
- ▶ **Calendar & Scheduling:** Timezone-aware slot scheduling with Google Calendar / CalDAV synchronization.

## Phase 3: Intelligence & Scale (65% → 100%)

- ▶ **AI Voice Mock Interviewer:** Integrate **Groq Whisper (STT)** + **ElevenLabs (TTS)** for realistic technical and behavioral voice interviews.
- ▶ **Semantic Vector Matching:** Deploy **PostgreSQL pgvector** and embeddings to match student skill gaps with ideal mentors using natural language.
- ▶ **Automated PDF Certificate Minting:** Cryptographically verified PDF certificates with QR codes for completed roadmaps and mentor accreditations.
- ▶ **Institutional Analytics:** Comprehensive college administrative dashboard tracking student progress and department metrics.

# Verification, Testing & Build Health Metrics

Zero-defect verification across compiler, linter, database, and production build.

**0**

## TYPESCRIPT ERRORS

Validated with `tsc --noEmit` under strict mode.

**0**

## ESLINT WARNINGS

Validated with `eslint --quiet` across all components.

**25 / 25**

## COMPILED ROUTES

Successfully built via Turbopack production compiler.

**100%**

## LIVE API PASS RATE

Tested Simple/Med/Hard & Mentor accreditation endpoints.

### Live API Endpoint Verification

- ▶ `POST /api/ai/assessment` verified for React (SIMPLE) and Python (HARD).
- ▶ `POST /api/mentor/assessment` verified: Score 92% correctly awarded Master Mentor (5.0★) and +15 XP.
- ▶ `GET /mentor/dashboard` and `GET /assessment` verified returning HTTP 200 OK.

### Security & Resilience Standards

- ▶ Stateless JWT with HttpOnly cookie flags and secure expiration.
- ▶ SQL Injection immunity via Prisma parameterized prepared statements.
- ▶ Graceful degradation ensuring tests run reliably whether AI key is present or offline.

## MILESTONE 1 DEFENSE SUMMARY

# Phase 1 Successfully Completed

The foundational 30% of SkillVerse is fully built, type-safe, and verified with live AI assessment and mentor accreditation. The platform is ready for live demonstration and prepared for Phase 2 WebRTC interactive rooms.



### Zero Placeholders

Working Database & Auth



### Real AI Pipeline

Groq LLaMA-3.3 70B



### Accredited Mentors

5.0★ Rating Algorithm